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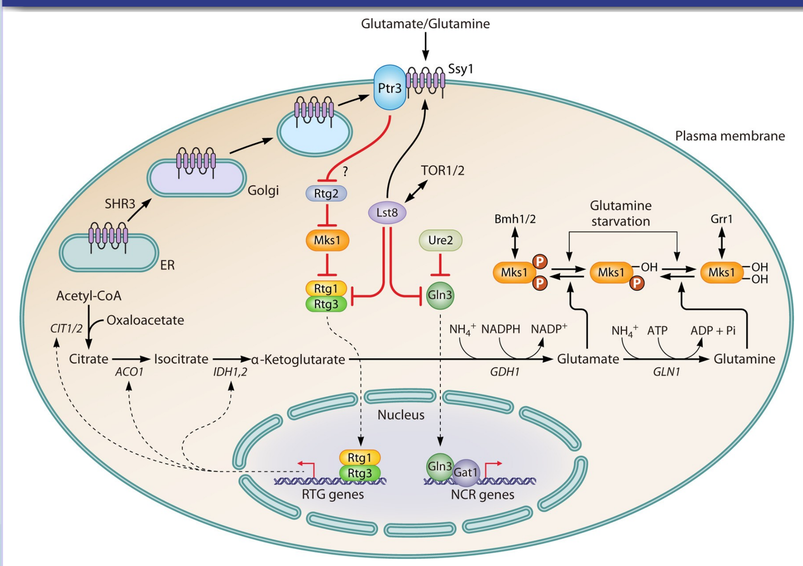
Predicting Yield:
Decoding Nitrogen Control of Yeast with Hybrid Mechanistic–ML Models

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1 · BACKGROUND



Nitrogen sensing and the NCR network in a yeast cell

Nitrogen decides whether yeast grows or produces.

Workhorse. *S. cerevisiae* makes recombinant proteins, enzymes and ethanol at industrial scale.

Sensing. SPS reads amino acids outside the cell, and TORC1 reads nitrogen status inside.

Switching. NCR (Gln3/Gat1 vs. Dal80/Deh1, gated by Ure2) uses preferred sources first.

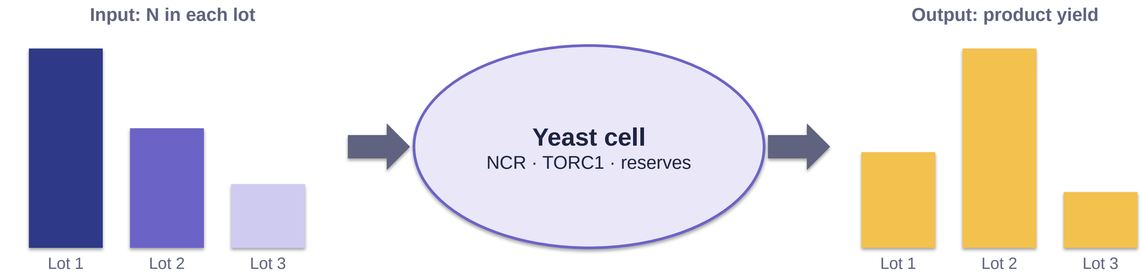
Reserves. Cells keep >50% spare metabolic and translational capacity to buffer nitrogen swings.

2 · THE PROBLEM

Varied yield: same strain, same process, different raw-material lot.

Raw-material nitrogen varies → yeast re-allocates resources → yield varies. No model predicts it.

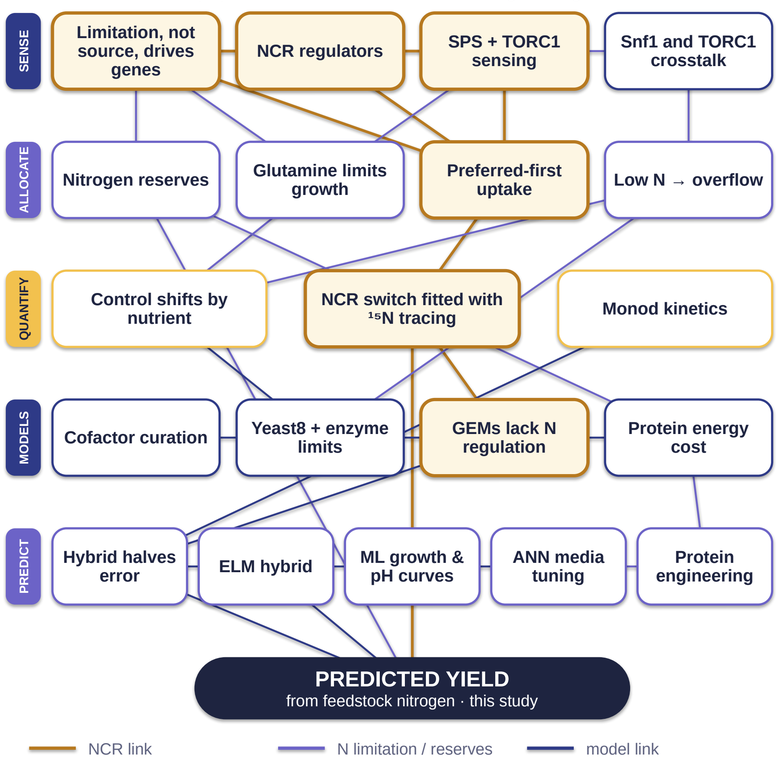
Lot = one supplier batch of the same raw material (e.g. yeast extract, peptone, molasses). Each lot carries a different amount and mix of nitrogen (total N, YAN, N forms).



Missing link: a model that predicts yield from feedstock nitrogen

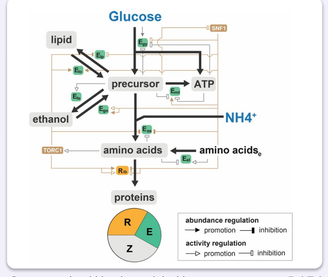
3 · NITROGEN KNOWLEDGE NETWORK

20 papers as one network, linked through nitrogen



4 · MODELLING APPROACHES: WHICH ONE CAN PREDICT YIELD?

1 KINETIC / MECHANISTIC (ODE)



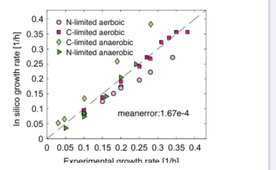
Coarse-grained kinetic model with proteome sectors R / E / Z

$\mu = \mu_{max}\, S / (K_S + S) \qquad dX/dt = \mu X$

- ✓ Meaningful parameters, little data needed
- ✓ Predicted a new run ($R^2 > 0.99$)
- ✗ One sugar, one N source only
- ✗ No NCR switching or reserves

Keywords: *Monod · mass balance*

2 GENOME-SCALE (GEM · FBA · FVA)



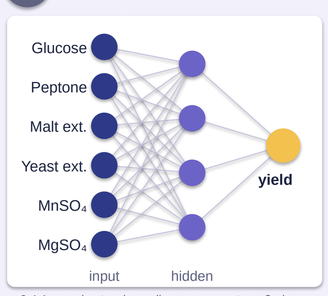
Yeast8: predicted vs. measured growth, C- and N-limited

$S \cdot v = 0 \qquad \max Z\ (\text{biomass}) \qquad \text{within flux bounds}$

- ✓ Whole network: 1,133 genes, 3,949 reactions
- ✓ FVA gives feasible flux ranges
- ✗ Steady state only, no NCR
- ✗ Overpredicts secreted protein ~1,000×

Keywords: *flux balance · enzyme limits*

3 DATA-DRIVEN ML (ANN · LSTM · SVR)



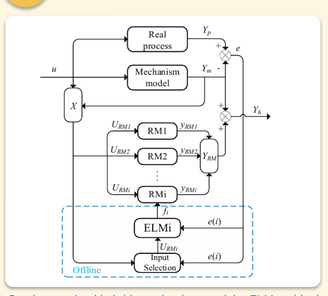
6-4-1 neural network: medium components → β-glucan yield

$f(x) = 1 / (1 + e^{-x}) \qquad \text{6-4-1 network}$

- ✓ Learns non-linear “sweet spots”
- ✓ 47 to 86% lower error than growth curves
- ✗ Black box, hard to trust
- ✗ Needs big data, extrapolates poorly

Keywords: *neural networks · back propagation*

4 HYBRID (MECHANISTIC + ML) ★ BEST FIT



Semi-supervised hybrid: mechanism model + ELM residual models

$Y_h = Y_m + Y_{RM} \qquad \text{(mechanism + residual model)}$

Takes care of:

- ✓ ① ② : ML learns missing regulation
- ✓ ③ : mechanism keeps it interpretable
- ✓ ~2× lower error on an unseen batch

Keywords: *residual learning · digital twin*

5 · OPEN GAPS

Gaps I can identify so far:

1. No model links feedstock nitrogen to yield.
2. Reserves and NCR are not linked to protein synthesis.
3. Data are mostly steady state, so dynamic nitrogen shifts are rarely modelled.
4. Mechanistic models drift, and ML alone is a black box.

6 · MY PROPOSED MODEL: HYPOTHESIS → TEST

*I expect a **hybrid of kinetic, GEM and ML models** to work best.*

INPUT Feedstock N — YAN · N forms · C:N → ① **Kinetic core** — Monod + φNCR(YAN) → ② **GEM / FVA** — + reserve limits → ③ **ML residual** — learns model error → **OUTPUT** Biomass, titre — + resistance score

HYPOTHESIS	Yield drops only when feedstock nitrogen pushes cells past their reserve capacity. Until then, NCR and reserves buffer the change.
PREDICTION	Lots where the simulated protein flux FVA width → 0 give robust, predictable titres, while wide ranges flag sensitive raw materials.
TEST	Train on public chemostat + N-pulse data → validate on held-out N sources → compare RMSE with kinetic-only and GEM-only models → wet-lab lots with varied YAN.

7 · IMPACT

- ▸ Screen raw materials **in silico**, before costly trials
- ▸ Fewer failed batches and consistent protein titres
- ▸ Cheaper, variable feedstocks become usable
- ▸ A reusable nitrogen aware modelling framework, with a review paper in preparation

8 · LOOKING FOR NEW APPROACHES

Your idea could be the missing piece!

Share your thoughts. I’m all ears and excited to learn from you.

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